

Malignant Infarction

Decompressive hemicraniectomy selection criteria, evidence, and supportive ICU care.

STAGE 1: 0 - 24 HOURS

Baseline Core & Serial NIHSS Checks

STAGE 2: 24 - 48 HOURS

Peak Edema Phase & Serial CT Scan

STAGE 3: < 48H SURGERY

Decompressive Hemicraniectomy

1. Decompressive Hemicraniectomy Selection Criteria

Clinical Deficit Severity:

- NIHSS > 15 (non-dominant hemisphere)
- NIHSS > 20 (dominant hemisphere)
- AND decrease in level of consciousness (NIHSS Item 1a score ≥ 1 / obtunded or stuporous)
- **Timing****: Surgery performed **within 48 hours** of onset.

Radiographic Markers:

- Infarction of $\geq 50\%$ of the MCA territory (CT/MRI)
- DWI core volume > 82 mL within 6 hours
- DWI core volume > 145 mL within 14 hours
- Midline shift or mass effect on repeat imaging
- **Surgical Spec****: Bone flap diameter $\geq 12-15$ cm with duraplasty.

2. Surgical Outcomes & Evidence (By Age Group)

Age < 60 Years (DECIMAL/DESTINY)

Surgery (22% Mort) vs Med (71% Mort)



Age < 60 Medical Control



Age ≥ 60 Years (DESTINY II)

Surgery (33% Mort) vs Med (70% Mort)



Age ≥ 60 Medical Control



■ mRS 0-3: Functional Independence / Mild-Mod Disability ■ mRS 4: Moderately Severe (Walk with assistance)
■ mRS 5: Severe Disability (Bedbound) ■ mRS 6: Death

• ****Age < 60****: NNT = 2 for survival, NNT = 4 for mRS ≤ 3 . | • ****Age ≥ 60 ****: NNT = 3 for survival, NNT = 25 for mRS ≤ 3 . *Goals-of-care discussion critical.

3. Supportive ICU Care & Medical Management

Positioning

Elevate HOB 30 degrees; maintain straight head/neck alignment to maximize venous outflow.

Fluids

Use Isotonic Saline (0.9% NS). **Avoid hypotonic fluids** (e.g. D5W, 0.45% NS, LR) which worsen edema. Maintain euvoolemia.

Osmotherapy

Consider **targeted PRN** hyperosmolar agents (HTS 3% or Mannitol) for acute decline or severe mass effect. *Prophylactic osmotherapy is not recommended.*

Metabolic

Target normothermia ($<37.8^{\circ}\text{C}$). Target normocapnia (PaCO₂ 35-45 mmHg); avoid hypoventilation/hypercapnia.

Steroids

Class III (Harmful): Corticosteroids are NOT recommended for reducing cerebral edema in acute ischemic stroke.

DECIMAL: Vahedi K et al. *Stroke*. 2007;38:2506-2517. | **DESTINY**: Jüttler E et al. *Stroke*. 2007;38:2518-2525.

HAMLET: Hofmeijer J et al. *Lancet Neurol*. 2009;8:326-333. | **DESTINY II**: Jüttler E et al. *N Engl J Med*. 2014;370:1091-1100.

AHA Guidelines: Wijdicks EF et al. *Stroke*. 2014;45:1222-1238.